Balu Nijampatnam

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Education

Indian Institute of Information Technology Jabalpur — **Jabalpur, Madhya Pradesh**
Bachelor of Technology in Computer Science and Engineering — CPI: 9.0 — *Aug. 2024 – Present*

Sri Chaitanya Junior College — **Vijayawada, Andhra Pradesh**
Class XII — 98.2% — *2022 – 2024*

Technical Skills

Languages: C++, JavaScript, Python, C
Frontend: React, HTML, CSS
Backend: Node.js, Express.js, REST APIs
Databases: MySQL, MongoDB
Machine Learning: Regression, Classification, Tree-Based & Ensemble Methods, Model Evaluation, EDA, Unsupervised Learning, Anomaly Detection
Libraries & Frameworks: Scikit-learn, openCV, NumPy, Pandas, Matplotlib, Seaborn
Developer Tools: Git, GitHub, VS Code, Postman, Vercel, Render
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Database Management Systems, Computer Networks

Projects

Crowd Management System | GitHub | Live Demo — **April 2026**
Python, OpenCV, Node.js, Express.js, scikit-learn, MongoDB, React

- Built a real-time crowd monitoring system using **YOLOv8** and **OpenCV**, combining frame skipping for computational efficiency with centroid-based tracking and dynamic grid-based density estimation, concurrently processing live CCTV streams and security-submitted field images.
- Designed and trained a **machine learning** pipeline that builds datasets from historical data, cross-validates candidate models, and predicts next-day crowd levels – deployed as an independent Flask microservice.
- Built **REST APIs** and a scheduled aggregation service (**Node.js**, **Express**, **MongoDB**) that automatically compresses raw detections into hourly/daily rollups for fast analytics queries.
- Built a role-based **React** dashboard (**Socket.IO**) with live stats, density heatmaps, prediction trends, and automated **Twilio SMS** alerts.

TradeX – Online Trading Exchange Simulator | GitHub | Live Demo — **June 2026**
C++, Node.js, Express.js, JavaScript, MongoDB, HTML, CSS

- Built a full-stack stock exchange simulator supporting user authentication, portfolio management, and real-time order placement through **REST APIs**.
- Implemented a **C++ price-time priority matching engine** supporting partial order execution, weighted-average portfolio tracking, and efficient order book management using **STL** data structures.
- Developed a responsive trading dashboard featuring live order books, portfolio analytics, profit/loss tracking, and seamless integration between the **Node.js** backend and C++ matching engine.

Smart Agriculture Decision Support System | GitHub — **August 2026**
Python, Flask, Scikit-learn, SHAP, RAG, LangGraph, OpenWeather API

- Developed a **location-aware crop recommendation engine** combining Random Forest predictions with soil, weather, and regional suitability factors for ranked crop recommendations.
- Added **yield, profitability, risk analysis, and SHAP explainability**, enabling the system to provide both crop recommendations and interpretable decision factors.
- Integrated a **domain-specific RAG pipeline and LangGraph agent** that retrieves agricultural knowledge and orchestrates weather, ML, and document-retrieval tools for context-aware recommendations.

Coding Profiles

LeetCode: balu_20 | 640+ Problems Solved | Rating: 1846 (Max: 1846)
CodeChef: balu_20 | DSA Rating: 1803 (Global Rank: 1011) | Rating: 1518 (Max: 1523)
Codeforces: balu_09 | Rating: 1190 (Max: 1190)

Achievements

- **Semi-Finalist**, Flipkart GRiD 8.0 – Software Development Track.
- Solved **1000+ algorithmic problems** across LeetCode,Codeforces, and CodeChef.
- Secured **Global Rank 2806** out of **48,162** participants in LeetCode Weekly Contest 514.
- Achieved **Rank 333** in CodeChef Starters 249.